

Efficacy and safety of abiraterone acetate plus prednisone and androgen deprivation therapy +/- docetaxel in older patients (≥ 70 years), with de novo metastatic-castration sensitive prostate cancer, compared to younger patients (< 70 years): The PEACE-1 trial.

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Background: Metastatic castration-sensitive prostate cancer (mCSPC) primarily affects older men (OM). In this post-hoc analysis we investigated the safety and efficacy of abiraterone acetate + prednisone (AAP) in older (≥ 70 years) and younger (< 70 years) patients in PEACE-1. **Methods:** Men with *de novo* mCSPC were allocated to standard of care (SOC), SOC + AAP, SOC + radiotherapy (RXT), or SOC + AAP + RXT in this 2x2 design phase 3 trial. SOC was initially androgen deprivation therapy (ADT) alone, then from Oct 2015 onwards, the use of docetaxel (D) was authorized as part of SOC (at the investigator's discretion until 2017, then, following the publication of LATITUDE and STAMPEDE trials, accrual was restricted to men receiving ADT+ D). Efficacy and safety in OM included in PEACE-1 were analyzed with the same methods used in the overall trial (Lancet 2022; 399: 1695-1707). **Results:** A total of 741 younger men (YM) (63%) and 431 OM (37%) were randomized. OM presented with more altered PS (ECOG 1-2) (36% vs 26% $p=0.0003$) and less frequent use of docetaxel (D) as part of SOC (66% vs 51% $p<0.0001$) than YM. Hypertension (56,5% vs 38,2%, $p<0,001$) and diabetes mellitus type 2 (15,5% vs 11%, $p=0,029$) were significantly more frequent in OM. Median time to AAP discontinuation was shorter (30.0 months [95%CI= 22.1; 35.4] vs 41.4 [95%CI= 31.5; 54.0] independently of D use and more frequently due to adverse events or death in the older than younger population. The benefit of AAP on radiographic progression-free survival (rPFS) tended to decrease with age in the overall population: (HR 0.65, 95%CI 0.42-1.01) in OM vs (HR 0.49, 95%CI 0.35-0.69) for YM. The same trend was observed on overall survival (OS): (HR 0.95, 95%CI 0.72-1.25) for OM vs (HR: 0.73, 95%CI 0.58-0.92) for YM. On the other hand, in men fit to receive the SOC composed of ADT+D, the rPFS benefit of AAP was comparable in OM (HR 0.55, 95%CI 0.29-1.04) and in YM (HR 0.5, 95%CI 0.33-0.78). The OS benefit of AAP was: (HR 0.80, 95%CI 0.53-1.2) and (HR 0.71, 95%CI 0.52-0.95) in OM and YM, respectively. Safety data show that severe adverse events (grade 3-5) were more frequent in OM receiving AAP in comparison with YM (69% vs 61%) while there was no difference between older and YM not receiving AAP (48% vs 47%). **Conclusions:** This post-hoc analysis of PEACE-1 suggests that, in the overall population, OM derive a lower benefit, both in terms of rPFS and OS, from adding AAP to SOC versus YM. This decreased benefit is likely due to more toxicity leading to more frequent and earlier drug discontinuation. Importantly, in OM fit enough to receive ADT+D, the benefit of adding AAP to SOC was comparable to YM. Clinical trial information: NCT01957436. Research Sponsor: French Government (grants PHRC-K-13-050 and PRT-K20-116); Janssen-Cilag, Ipsen, Sanofi.